Lenox Mucumbi

Backend Developer | TypeScript/Node.js
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EDUCATION

Dom Bosco Institute of Higher Education Maputo, Mozambique
Bachelor of Science in Computer Programming *Mar. 2023 – Expected 2027*

EXPERIENCE

ITC Trainee Instructor | Web Development Sep. 2026 – Present

- Started teaching the module Developing Web Applications with JavaScript.
- Currently teaching Using JavaScript Libraries, covering practical use of JavaScript libraries in web development.

STEMMOZ | Robotics & Programming Mar. 2026 – Jul. 2026

STEM Education and Technical Support

- Assembled and programmed educational robots using Arduino, Avishkaar, and REV Robotics platforms.
- Supported programming and robotics teaching activities and provided technical assistance during technology events.
- Contributed to an AI solution for real-time posture analysis at the VisionHer event.

Clardia Tours | Front-End Developer Sep. 2025 – Dec. 2025

Tourism platform | clardiatours.com

- Developed and maintained responsive user interfaces for a tourism platform.
- Built reusable tour-package showcase components that render structured JSON data for 9 tour packages, each with 12 pages, improving maintainability.
- Applied responsive design and UI performance practices to improve the user experience.
- Continue to deliver platform updates on request from the site owner.

PROJECTS

Stock Management System | STEMMOZ Jul. 2026 – Present

NestJS, TypeScript, PostgreSQL, Supabase, Next.js, Docker

- Developed a stock-management software with role-based access for administrators, stock managers, and operators.
- Designed relational data structures for inventory management, incorporating DB optimisation techniques such as indexing and query refinement to ensure high-performance data retrieval.
- Integrated PostgreSQL through Supabase and prepared Docker-based containerized deployment and CI/CD workflows.
- Conducted comprehensive DB optimisation by analyzing execution plans and restructuring schema relationships to minimize latency in complex inventory queries.

BookStore Microservices | E-commerce Platform Jul. 2026 – Present

NestJS, TypeScript, Sequelize, MySQL, RabbitMQ, Redis

- Designing a distributed e-commerce architecture for bookstore management using decoupled Node.js/TypeScript NestJS microservices.
- Implemented event-driven communication between decoupled NestJS microservices using RabbitMQ, leveraging Redis for high-performance caching to reduce database latency and improve system throughput.
- Modeling relational data with Sequelize and MySQL across independent services.
- Optimized system performance by implementing multi-level caching strategies for frequently accessed product and user session data to ensure high availability and scalability.

SMS Spam Classifier | AI Learning Agent Oct. 2025 – Dec. 2025

Python, Scikit-Learn, Pandas, Naive Bayes, Joblib, NLP

- Built an intelligent spam/ham SMS classifier using Multinomial Naive Bayes that improves its performance over time.
- Implemented an NLP pipeline to clean raw text and perform efficient data parsing and term frequency extraction using Pandas and Scikit-Learn.
- Persisted trained probability models with Joblib to support real-time inference.
- Developed custom data parsing scripts to transform unstructured SMS datasets into structured formats for model training and inference.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, SQL
Web Technologies: CSS
Runtime & Frameworks: Node.js, React, Next.js, Express, NestJS
Databases: PostgreSQL, MySQL, MongoDB, Redis
Developer Tools: Docker, Git, GitHub, CI/CD, Supabase, VS Code, RabbitMQ
Methodologies: Scrum, Agile, Clean Code, Waterfall

LANGUAGES

Portuguese(fluent) | Xichangana: Not specified | English: Intermediate (B1)